

Omar Ashraf

Machine Learning & Full-Stack Engineering Intern Candidate

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SUMMARY

Final-year Communications and Electronics Engineering student with hands-on experience building and deploying ML/NLP systems (RAG, transformer fine-tuning, computer vision) and full-stack applications. Vice Head of an IEEE ML team who has taught a computer vision course. Strong self-directed learner seeking an internship to apply ML and backend skills on real-world products.

EDUCATION

B.Sc. Communications and Electronics Engineering

Expected 2027

Faculty of Engineering, Alexandria University

Self-study: completed the OSSU Computer Science curriculum; linear algebra (Strang), probability (Stat 110), backpropagation from scratch, CNNs, transformers, BERT.

EXPERIENCE & LEADERSHIP

Vice Head, ML Team — IEEE SSCS Student Branch

Nov 2025 – Present

- Help lead the ML team of the IEEE Solid-State Circuits Society student branch at Alexandria University.
- Taught an 8-week computer vision course to the ML team.

Programming Tutor (Kids)

Freelance

- Teach programming fundamentals to children.

PROJECTS

RECALL — RAG Chatbot

LangChain, FAISS, FastAPI, JavaScript

- Built a retrieval-augmented chatbot with a LangChain (LCEL) pipeline, FAISS vector search, a FastAPI backend and a vanilla JS frontend using OpenRouter LLMs.

Toxic Content Classifier & Moderation Pipeline

PyTorch, Hugging Face, FastAPI

- Trained a 15-class toxicity classifier on BeaverTails: Word2Vec + BiLSTM baseline, then DistilBERT fine-tuned with LoRA; deployed through FastAPI.
- Built a zero-shot content moderation pipeline with Llama Guard (NLP course project at Neurova).

Personal Finance Manager

FastAPI, PostgreSQL, React PWA, Docker

- Full-stack finance tracking app with a FastAPI + PostgreSQL backend and a React progressive web app, orchestrated with Docker Compose.

Soundband — Assistive Device for the Deaf

ESP, MFCC, Embedded ML

- Built a working audio-to-text device for an IEEE Arduino competition: on-device MFCC feature extraction and a self-trained model deployed to an ESP microcontroller.

HackerMode

PyQt6, Python

- Windows lock screen that gates access behind daily tasks, with Anki integration.

SKILLS

ML / AI: PyTorch, Hugging Face Transformers, LoRA fine-tuning, CNNs, object detection (FPN), RAG, LangChain, FAISS

Backend & Web: Python, FastAPI, PostgreSQL, JWT auth, React, Tailwind CSS, JavaScript, REST APIs

Tools: Docker & Docker Compose, Git, Linux/WSL, VS Code; networking fundamentals (TCP/IP, DNS)

Other: Embedded systems (ESP, Arduino), PyQt6, computer architecture fundamentals

LANGUAGES

Arabic (native) | English (fluent) | Japanese (basic)